

Kalina Hristova Petrova

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Nationality: Bulgarian | Birth date: 13.10.1994

EDUCATION

- 09.2019 – 05.2024 **Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland**
PhD in Theoretical Computer Science
Advisor: Prof. Dr. Angelika Steger
- 09.2017 – 06.2019 **Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland**
Master of Science in Computer Science, track: Theoretical Computer Science
GPA: 5.91 out of 6.00
- 09.2013 – 06.2017 **Princeton University, Princeton, New Jersey, USA**
Bachelor of Science in Engineering in Computer Science
GPA: 3.93 out of 4.00, summa cum laude

WORK EXPERIENCE

- 06.2024 – present **Postdoctoral researcher, Institute of Science and Technology Austria (ISTA), Klosterneuburg, Austria**
Marie Skłodowska-Curie fellow under the IST-BRIDGE program, 06.2024 – 05.2026
 - Conducted research in probabilistic and extremal combinatorics under the mentorship of Prof. Dr. Matthew Kwan
- 05.2019 – 08.2019 **Software Engineering Intern, Google, Zurich, Switzerland**
 - Generated datasets for evaluating machine learning models for the Google Ads team
 - Analyzed the results of this evaluation and provided recommendations for the team's future work
- 07.2014 – 08.2014 **Student Mentor/Lecturer, Summer Research School in Mathematics and Informatics, Bulgaria**
 - Mentored three student research projects in Computer Science
 - Gave a lecture to the Bulgarian extended national team for the International Olympiad in Informatics

TEACHING

- 12.2024 – 01.2025 Lecturer for **Topics in Ramsey theory**, Institute of Science and Technology Austria (ISTA)
- 09.2023 – 12.2023 Creator of exercises for **Advanced algorithms**, ETH Zurich
- 03.2020 – 05.2024 Teaching assistant for **Algorithms and probability**, ETH Zurich (5 times)
- 03.2020 – 05.2024 Supervisor for **Seminar on randomized algorithms and probabilistic methods**, ETH Zurich (4 times)
- 09.2019 – 02.2024 Teaching assistant for **Algorithms lab**, ETH Zurich (5 times)
- 09.2019 – 02.2023 Exam grader for **Randomized algorithms and probabilistic methods**, ETH Zurich (3 times)
- 02.2017 – 07.2017 Creator of lecture notes for **Linear locally decodable codes**, Princeton University

STUDENT SUPERVISION

- 02.2024 – 05.2024 Jonas Lill, Bachelor thesis, "Parameterized Algorithm for Max-Cut in Multigraphs", ETH Zurich
- 09.2023 – 12.2023 Andor Vári-Kakas, semester project on size-Ramsey numbers of hypertrees, ETH Zurich
- 02.2023 – 06.2023 Patryk Morawski, semester project, "Randomly perturbed digraphs also have bounded-degree spanning trees", ETH Zurich
- 04.2022 – 10.2022 Christopher Burckhardt, Bachelor thesis, "Analysis of weight bounded and restricted non-preemptive tree packing", ETH Zurich
- 09.2020 – 03.2021 Erik Jahn, Master thesis, "Sample-efficient learning with neural tuning curves", ETH Zurich
- 12.2019 – 06.2020 Nina Laura Corvelo Benz, Master thesis, "Using inhibitory signals for error encodings in networks with tuning curves", ETH Zurich
- 10.2019 – 04.2020 Robert Meier, Master thesis, "Immediate plasticity and adaptive tuning yield optimal sample efficiency in learning", ETH Zurich

ACTIVITIES AND OUTREACH

- 08.2026 Co-organizer of the First Bulgarian Workshop in Probabilistic and Extremal Combinatorics (BWPEC), together with Lyuben Lichev and Simona Boyadzhyska, in Oryahovitsa, Bulgaria

08.2026	Volunteer lecturer at a combinatorics summer camp for high school students in Oryahovitsa, Bulgaria
10.2018 – 05.2024	Volunteer lecturer as part of the Swiss Olympiad in Informatics
06.2021	Guide at the European Girls' Olympiad in Informatics
10.2016 – 05.2017	Volunteer teacher, Mathematics for underprivileged high school students, STEMCivics Purplefect Palace High School, Ewing Township, New Jersey, USA
09.2012 – 05.2018	Mentor of two high-school students for their research projects for High School Student Institute of Mathematics and Informatics conferences in Bulgaria, among which • On a special case of the bin packing problem • Computing unsigned reversal distance between genetic sequences using genetic algorithms • A new genetic algorithm for the 3-dimensional matching problem

AWARDS AND DISTINCTIONS

2024	Marie Skłodowska-Curie fellow under the IST-BRIDGE program, Institute of Science and Technology Austria (ISTA)
2017	Recipient of Excellence Scholarship & Opportunity Programme (ESOP), ETH Zurich
2017	Sigma Xi Award for Outstanding Undergraduate Researcher for research done at Princeton University
2017	Participation in the ACM Intercollegiate Programming Contest, World Finals 2017 in Rapid City, USA
2017	Honorable Mention for the Computing Research Association's Outstanding Undergraduate Researcher Award for research done at Princeton University
2017	Elected to membership in the Society of Sigma Xi: The Scientific Research Honor Society
2017	Elected to membership in the Phi Beta Kappa Society for academic excellence
2016	1st place in a team of 3 out of 48 teams at the Regional ACM International Collegiate Programming Contest 2016, Greater New York region, New York, USA
2016	Winner of the Accenture Prize for Academic Excellence in Computer Science at Princeton University
2015	Elected to membership in the Tau Beta Pi National Engineering Society, USA
2015	Awarded the Shapiro Prize for Academic Excellence at Princeton University
2015	2nd place in a team of 3 out of 49 teams at the Regional ACM International Collegiate Programming Contest 2015, Greater New York region, New York, USA

LANGUAGES

Bulgarian	Native	German	B2
English	Fluent	French	B1

TALKS

Invited conference/workshop talks

07.2026	31st British Combinatorial Conference (BCC 2026), Cardiff University — Probabilistic Combinatorics Mini-Symposium
07.2025	Extremal and Probabilistic Combinatorics Workshop, International Centre for Mathematical Sciences (ICMS), Edinburgh
03.2024	A Spring Day of Combinatorics, University of Birmingham

Invited seminars

09.2026	Algorithms, Dynamics, and Information Flow in Networks Seminar, DFG Research Unit
06.2026	Discrete Mathematics Working Group (AGDM) Seminar, University of Vienna and TU Wien
04.2026	(Not So) Informal Probability Seminar, University of Vienna and TU Wien
11.2025	Atlantic Graph Theory Seminar
05.2025	Discrete Mathematics Seminar, University of Passau
12.2024	Graz Combinatorics Seminar, TU Graz
09.2024	Combinatorial Mathematics Society of Australasia Seminar
05.2024	Theoretical Computer Science & Discrete Mathematics Research Seminar, University of Heidelberg
10.2022	Seminar on Combinatorics, Games and Optimisation, London School of Economics

Internal seminars

- 12.2025 Combinatorics, Geometry and Topology (CGT) Seminar, Institute of Science and Technology Austria (ISTA)
- 09.2021 – 12.2025 Mittagseminar, ETH Zurich, 8 talks in total

Contributed conference/workshop talks

- 09.2025 Discrete Mathematics Meeting 2025, TU Graz
- 08.2025 Random Structures & Algorithms 2025, TU Wien
- 11.2024 DIMEA Combinatorial Potluck 2024, Masaryk University
- 09.2023 Constructive and Probabilistic Methods in Combinatorics Workshop, University of Zagreb
- 06.2023 Random Structures & Algorithms 2023, Carnegie Mellon University
- 08.2022 Random Structures & Algorithms 2022, Institute of European Culture in Gniezno
- 07.2019 International Colloquium on Automata, Languages and Programming (ICALP 2019), University of Patras

RESEARCH INTERESTS

- Probabilistic and extremal combinatorics
- Combinatorial design theory
- Random graphs
- Algorithms and data structures
- Ramsey theory
- Randomized algorithms

JOURNAL PUBLICATIONS

- R. Montgomery, K. Petrova, A. Ranganathan, and J. Tan, “Packing subdivisions into regular graphs”, *Proceedings of the American Mathematical Society* 154 (2026), 3693–3704.
- N. Draganić, M. Kaufmann, D. Munhá Correia, K. Petrova, and R. Steiner, “Size-Ramsey numbers of structurally sparse graphs”, *Random Structures & Algorithms* 68(2) (2026), e70059.
- P. Morawski and K. Petrova, “Randomly perturbed digraphs also have bounded-degree spanning trees”, *The Electronic Journal of Combinatorics* 33(2) (2026), P2.24.
- M. Christoph, R. Nenadov, and K. Petrova, “The Hamilton space of pseudorandom graphs”, *Journal of Combinatorial Theory, Series B* 176 (2026), 254–267.
- S. Boyadzhyska, S. Das, T. Lesgourgues, and K. Petrova, “Odd-Ramsey numbers of complete bipartite graphs”, *European Journal of Combinatorics* 131 (2026), 104235.
- N. Draganić and K. Petrova, “Size-Ramsey numbers of graphs with maximum degree three”, *Journal of the London Mathematical Society* III(3) (2025), e70116.
- J. Lill, K. Petrova, and S. Weber, “Linear-time MaxCut in multigraphs parameterized above the Poljak-Turzík bound”, *Algorithmica* 87 (2025), 983–1007.
- M. Christoph, K. Petrova, and R. Steiner, “A note on digraph splitting”, *Combinatorics, Probability and Computing* 34(4) (2025), 559–564.
- Y. Pehova and K. Petrova, “Embedding loose spanning trees in 3-uniform hypergraphs”, *Journal of Combinatorial Theory, Series B* 168 (2024), 47–67.
- J. Lengler, A. Martinsson, K. Petrova, P. Schnider, R. Steiner, S. Weber, and E. Welzl, “On connectivity in random graph models with limited dependencies”, *Random Structures & Algorithms* 65(2) (2024), 411–448.
- K. Petrova and M. Trujić, “Transference for loose Hamilton cycles in random 3-uniform hypergraphs”, *Random Structures & Algorithms* 65(2) (2024), 313–341.

CONFERENCE PUBLICATIONS

- J. Lill, K. Petrova, and S. Weber, “Linear-time MaxCut in multigraphs parameterized above the Poljak-Turzík bound”, 19th International Symposium on Parameterized and Exact Computation (IPEC 2024), Schloss Dagstuhl–Leibniz-Zentrum für Informatik, LIPIcs 321 (2024), 2:1–2:19.
- K. Lakis, J. Lengler, K. Petrova, and L. Schiller, “Improved bounds for graph distances in scale free percolation and related models”, *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2024)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, LIPIcs 317 (2024), 74:1–74:22.
- J. Lengler, A. Martinsson, K. Petrova, P. Schnider, R. Steiner, S. Weber, and E. Welzl, “On connectivity in random graph mod-

els with limited dependencies”, *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM 2023)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, LIPIcs 275 (2023), 30:1–30:22.

- Y. Pehova and K. Petrova, “Minimum vertex degree conditions for loose spanning trees in 3-graphs”, *European Conference on Combinatorics, Graph Theory and Applications 2023*, 754–759.
- B. Haeupler, F. Kuhn, A. Martinsson, K. Petrova, and P. Pfister, “Optimal strategies for patrolling fences”, *Proceedings of the 46th International Colloquium on Automata, Languages, and Programming (ICALP 2019)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, LIPIcs 132 (2019), 144:1–144:13.
- K. Petrova and R. Tarjan, “A dynamic data structure for segment intersection queries: Extended abstract”, *Fifteenth International Workshop on Algebraic and Combinatorial Coding Theory (ACCT 2016)*, 2016, 244–249.

PREPRINTS

- J. Allsop, C. Bowtell, T. Lesgourgues, K. Petrova, “Improved bounds on completion of partial Latin squares”, September 2026.
- S. Antoniuk, A. E. Díaz, K. Petrova, M. Stojaković, “On constructing small subgraphs in the budget-constrained random graph process”, February 2026.
- S. Cambie, A. Freschi, P. Morawski, K. Petrova, A. Pokrovskiy, “Ramsey number of a cycle versus a graph of a given size”, January 2026.
- S. Boyadzhyska, S. Das, T. Lesgourgues, and K. Petrova, “Odd-Ramsey numbers of Hamilton cycles”, November 2025.
- M. Christoph, B. Janzer, K. Petrova, R. Steiner, “Extending Thomassen’s conjecture to directed graphs”, October 2025.
- M. Kwan, K. Petrova, and M. Sawhney, “Parities in random Latin squares”, September 2025.
- T. Makai, M. Pasch, K. Petrova, and L. Schiller, “Sharp thresholds for higher powers of Hamilton cycles in random graphs”, February 2025.

THESES

- “Embedding Large Structures in Adversarially Modified Graphs and Hypergraphs” – Doctoral thesis, ETH Zurich, 2024, under the supervision of Prof. Dr. Angelika Steger.
- “Parallel variational autoencoders for image segmentation” – Master thesis, ETH Zurich, 2019, under the supervision of Prof. Dr. Angelika Steger, Asier Mujika, and Frederik Benzing.
- “A dynamic data structure for segment intersection queries” – Bachelor independent work, Princeton University, 2016, under the supervision of Prof. Dr. Robert Tarjan.